CONFIGURE IPv6 ADDRESSING

1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Ans:-

Operating System: Windows 10/11

* **Step 1**: Open Command Prompt as Administrator.

  Press `Win + X` and select Terminal (Admin) or Command Prompt (Admin).

* **Step 2:** Check your network adapter name.

  Type: `netsh interface ipv6 show interfaces`

Find your active network adapter name, such as `Ethernet` or `Wi-Fi`.

* **Step 3:** Assign a static IPv6 address.

  For example, if your adapter is named `Ethernet`,

Type: `netsh interface ipv6 set address interface="Ethernet" address=2001:db8:1:: 10`

This assigns the example IPv6 address `2001:db8:1:: 10` to the adapter.

* **Step 4:** Verify the configuration.

  Type: `ipconfig`

  Look for the IPv6 Address under your Ethernet adapter.

* **You can also use**: `netsh interface ipv6 show addresses`

Expected result:

```
Ethernet adapter Ethernet:

   IPv6 Address . . . . . : 2001:db8:1:: 10
```

2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

   Ans:-

System Used: Windows

Assign a Unique Local IPv6 Address

* **Open Command Prompt as Administrator and run**: netsh interface ipv6 set address name="Wi-Fi" address=fd00:1:1::10

Configure IPv6 Address on another LAN Device

Example address for the second device: fd00:1:1:: 10

Test IPv6 Connectivity From the first computer, run: ping -6 fd00:1:1:: 10

The output should show:

Reply from fd00:1:1:: 20: time<1ms

Reply from fd00:1:1:: 20: time<1ms

Result: The unique local IPv6 address fd00:1:1:: 10 was configured successfully, and connectivity with the LAN device fd00:1:1:: 20 was verified using ping -6.

3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.

Ans:-

System Used: Windows

Enable Temporary IPv6 Addresses

Open Command Prompt as Administrator and run: netsh interface ipv6 set privacy state=enabled

Check the Current IPv6 Address Run: Ipconfig

Example output:

Temporary IPv6 Address: 2001:db8:1:: abcd

Disconnect and Reconnect the Network Disconnect the Wi-Fi, reconnect it, and run: Ipconfig

Check the Temporary IPv6 Address again.

Restart the Computer after restarting, run: Ipconfig

Check whether the temporary IPv6 address has changed.

* **Result:** The temporary IPv6 address was configured and checked before and after network reconnection/reboot. The observed address can change because temporary IPv6 addresses are designed to provide privacy.

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Ans: - **Language Used**: Python

```
# Zomato-Style Restaurant Search Using IPv6

Restaurants = [
   {"name": "Pizza Palace", "ipv6": "fd12:3456::10"},
   {"name": "Burger House", "ipv6": "fd12:3456::20"},
   {"name": "Spice Garden", "ipv6": "fd12:3456::30"},
   {"name": "Royal Restaurant", "ipv6": "fd12:3456::40"}
]

Search = input ("Enter restaurant name: ").lower ()

Found = False

For restaurant in restaurants:
   If search in restaurant ["name"].lower ():
      Print ("Restaurant:” restaurant ["name"])
      Print ("IPv6 Address:” restaurant ["ipv6"])
      Found = True

If not found:
   Print ("No matching restaurant found.")
```

Example output

Program output

Enter restaurant name: pizza Restaurant: Pizza Palace IPv6 Address: fd12:3456::10

Conclusion

The program searches restaurant names using partial-name matching and displays the matching restaurant's IPv6 address. It simulates a basic restaurant search system like Zomato.

5. Use ChatGPT or Copilot to generate a sample IPv6 Subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Ans:-

IPv6 Subnetting Scenario and Configuration

Operating System: Windows 10/11
Network Adapter: WiFi

1. AI-Generated IPv6 Subnetting Scenario

* A small organization has received the documentation IPv6 prefix `2001:db8:1200::/48`. The organization needs three subnets for different departments.

Subnet 1 – Administration

- Network Prefix: `2001:db8:1200:1::/64`
- Example IPv6 Address: `2001:db8:1200:1::10`

Subnet 2 – Sales

- Network Prefix: `2001:db8:1200:2::/64`
- Example IPv6 Address: `2001:db8:1200:2::10`

Subnet 3 – IT Department

- Network Prefix: `2001:db8:1200:3::/64`
- Example IPv6 Address: `2001:db8:1200:3::10`

Each subnet uses a `/64` prefix, which is commonly used for IPv6 LANs.

Note: The `2001:db8:: /32` range is reserved for documentation and practice. It is not intended for real internet connectivity.

2. Configuration Steps on Windows

* **Step 1: Open PowerShell as Administrator**

1. Press the Windows key.
2. Search for PowerShell.
3. Right-click it and select **Run as administrator**.
4. Click yes if prompted.

* **Step 2: Check the network adapter**

Run:

```
netsh interface ipv6 show interfaces
```

The active adapter in this example is `WiFi`.

* **Step 3: Assign an IPv6 address from Subnet 1**

Run:

```
netsh interface ipv6 set address interface="WiFi" address=2001:db8:1200:1::10
```

* **Step 4: Verify the IPv6 configuration**

Run: `ipconfig`

Look under **Wireless LAN adapter WiFi** for the IPv6 Address field.

You can also run: `netsh interface ipv6 show addresses interface="WiFi"`

Expected example address:

```
2001:db8:1200:1:: 10
```

3. Important Notes

- Administrator privileges are required to change the IPv6 address.
- This is a documentation-only address, so it is suitable for a classroom demonstration but not normal internet use.
- Assigning the address does not automatically configure the router or guarantee connectivity.
- For a real network, use an IPv6 prefix provided by your router, ISP, or network administrator.
- If Windows reports an error, confirm that PowerShell is running as Administrator and that the adapter is named `WiFi`.

4. Conclusion

I created an IPv6 Subnetting scenario containing three `/64` subnets and documented the Windows command-line steps to assign and verify an IPv6 address on the WiFi adapter.